

MTH 202 Chapter 1: Three-Dimensional Space & Vectors

1.5 Parametric Equations of Lines

Question: In Algebra 1 what information do you need in order to find the equation of a line?

Two points on the line OR A point on the line and the slope of the line

Motivation: What information do we need to determine the equation of a line in 3-dimensions and higher?

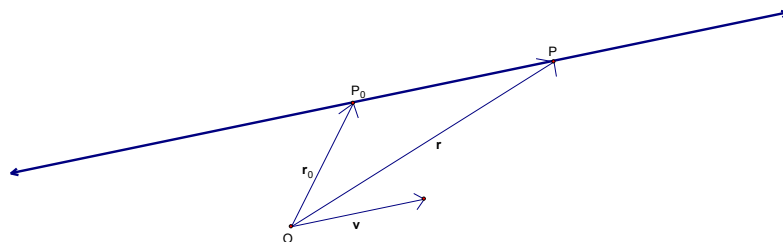
We can think of identifying a line in 3-space if we know:

- A point on the line (call it P_0)
- A direction in which the line “points”

Definition: If a line, l , is parallel to a vector $\mathbf{v}$, then we call $\mathbf{v}$ a direction vector for l .

In the picture we can see, O is the origin, P is an arbitrary point on the line, and $\mathbf{v}$ is our direction vector.

We can create vectors $\mathbf{r}$ & $\mathbf{r}_0$ and should immediately notice that $\mathbf{r}$ is the sum of $\mathbf{r}_0$ and some scalar multiple of $\mathbf{v}$.



We then develop the following terminology.

Terminology: The vector equation of a line L is $\mathbf{r} = \mathbf{r}_0 + t\mathbf{v}$ where $\mathbf{r}_0$ is the position vector of a point on the line and $\mathbf{v}$ is a vector parallel to the line. Each real number value of t (the parameter) gives the position vector of a point on L .

Note: This idea of a vector equation for a line works for two dimensions and higher.

Terminology: In 3 dimensions if we let $\mathbf{r} = \langle x, y, z \rangle$, $\mathbf{r}_0 = \langle x_0, y_0, z_0 \rangle$, and $\mathbf{v} = \langle a, b, c \rangle$, the vector equation of the line L becomes: $\langle x, y, z \rangle = \langle x_0, y_0, z_0 \rangle + t\langle a, b, c \rangle \Rightarrow \langle x, y, z \rangle = \langle x_0 + at, y_0 + bt, z_0 + ct \rangle$.

This leads to other ways of representing the equation of a line. For example, we could represent the line L as three separate equations by equating components of the vectors on each side of the vector equation. In doing

this we obtain:
$$\begin{cases} x = x_0 + at \\ y = y_0 + bt \\ z = z_0 + ct \end{cases}$$
 These equations are called the parametric equations of the line L (or a parameterization of L).

Example 1: Find the vector equation and parametric equations of the line that passes through the point $(2, -3, 4)$ and is parallel to the vector $-\mathbf{i} + \mathbf{j} + \mathbf{k}$.

We see that $\mathbf{r}_0 = \langle 2, -3, 4 \rangle$ and $\mathbf{v} = \langle -1, 1, 1 \rangle$. Thus, the vector equation is $\langle x, y, z \rangle = \langle 2, -3, 4 \rangle + t \langle -1, 1, 1 \rangle$.

The parametric equations are: $x = 2 - t$ $y = -3 + t$ $z = 4 + t$.

Note: The instructor will show a 3 dimensional graph.

Example 2:

a) Find parametric equations for the line that passes through the points $(2, -3, 4)$ and $(4, 2, 1)$.

If we name the two given points as A and B respectively, then we know that a vector parallel to the desired line is: $\mathbf{v} = \overrightarrow{AB} = \langle 4 - 2, 2 - (-3), 1 - 4 \rangle = \langle 2, 5, -3 \rangle$.

Thus, a vector equation for the line is $\langle x, y, z \rangle = \langle 2, -3, 4 \rangle + t \langle 2, 5, -3 \rangle$

and parametric equations for the line are:
$$\begin{cases} x = 2 + 2t \\ y = -3 + 5t \\ z = 4 - 3t \end{cases}$$

b) Find the point at which the line that you found in (a) intersects the xz -plane.

We know that any point in the xz -plane has 0 as its y -coordinate.

So, we need to know what value(s) of t causes y to be 0.

So, we solve: $0 = -3 + 5t \Rightarrow t = \frac{3}{5}$.

Then, we plug $t = \frac{3}{5}$ into the equations determining x and z , and we find $x = \frac{16}{5}$ and $z = \frac{11}{5}$.

Thus, the line intersects the xz -plane at $\left(\frac{16}{5}, 0, \frac{11}{5}\right)$.

c) What restrictions should you place on t if you want your parametric equations from (a) to determine only the line segment connecting $(2, -3, 4)$ and $(4, 2, 1)$?

We note that when $t = 0$ we get the point $(2, -3, 4)$ and when $t = 1$ we get the point $(4, 2, 1)$. So, the parametric

equations
$$\begin{cases} x = 2 + 2t \\ y = -3 + 5t \\ z = 4 - 3t \end{cases}$$
 where $0 \leq t \leq 1$ determine the line segment from $(2, -3, 4)$ to $(4, 2, 1)$.

Note: Part (c) drives home the idea that parametric equations may be a good way to describe the movement of a particle in 3 space.

Remark: If two lines have parallel direction vectors, the two lines are parallel. If two lines intersect at a point we say that the two lines intersect. If two lines do not intersect and are not parallel we say that the lines are skew.

Example 3: Suppose the line L_1 is given by: $\langle x, y, z \rangle = \langle -3, 0, 1 \rangle + t \langle 1, 2, 1 \rangle$, and the line L_2 is given by: $\langle x, y, z \rangle = \langle 1, 4, -7 \rangle + s \langle -1, -1, 2 \rangle$. Determine if these lines are parallel, intersecting, or skew.

We notice these lines are not parallel because $\langle 1, 2, 1 \rangle$ and $\langle -1, -1, 2 \rangle$ are not scalar multiples of each other.

To determine if the lines are intersecting we must determine if there exist real numbers s and t so that:

$$\langle 1, 4, -7 \rangle + s \langle -1, -1, 2 \rangle = \langle -3, 0, 1 \rangle + t \langle 1, 2, 1 \rangle \Rightarrow \langle 1-s, 4-s, -7+2s \rangle = \langle -3+t, 2t, 1+t \rangle.$$

We could also look at this through the lens of parametric equations of these lines:

$$L_1 \text{ is given by } \begin{cases} x = -3 + t \\ y = 2t \\ z = 1 + t \end{cases} \text{ and } L_2 \text{ is given by } \begin{cases} x = 1 - s \\ y = 4 - s \\ z = -7 + 2s \end{cases}.$$

$$\text{Therefore, we want values of } s \text{ and } t \text{ such that } \begin{cases} 1-s = -3+t \\ 4-s = 2t \\ -7+2s = 1+t \end{cases}.$$

Equating first components we have: $1-s = -3+t \Rightarrow t = 4-s$.

If we use this fact and equate the second components from the vector equation we have

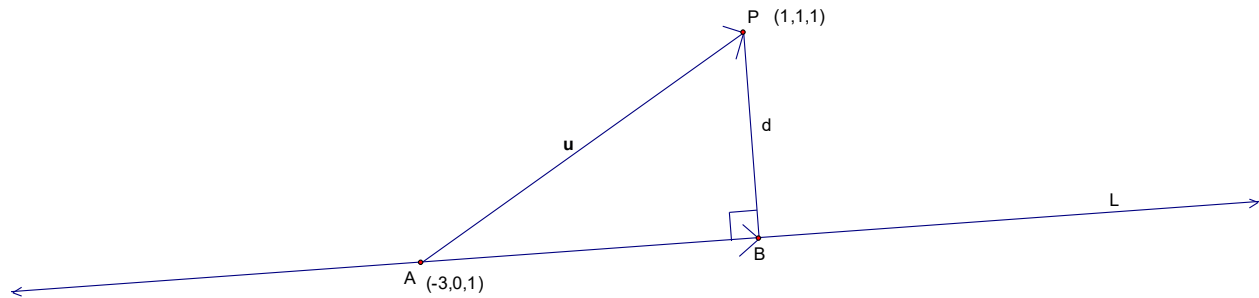
$$4-s = 2t \Rightarrow 4-s = 2(4-s) \Rightarrow 4-s = 8-2s \Rightarrow s = 4 \text{ and } t = 0.$$

We note that if $s = 4$ and $t = 0$, the vector equation becomes $\langle -3, 0, 1 \rangle = \langle -3, 0, 1 \rangle$. Thus, the lines do in fact intersect at the point $(-3, 0, 1)$.

Note: If the lines were skew the third component would not have matched after we found s and t .

Example 4: Suppose the line L is given by: $\langle x, y, z \rangle = \langle -3, 0, 1 \rangle + t \langle 1, 2, 1 \rangle$. Find the exact distance between this line and the point P which has coordinates $(1, 1, 1)$.

We note that $(-3, 0, 1)$ is a point on the line. A picture of the situation is:



We wish to find the length d . Our strategy will be to find $\|\mathbf{u}\|$ and $\|\overrightarrow{AB}\|$, and then use the Pythagorean theorem (alternatively one could try to find $\|\mathbf{u} - \overrightarrow{AB}\|$).

We know that $\mathbf{u} = \langle 1 - (-3), 1 - 0, 1 - 1 \rangle = \langle 4, 1, 0 \rangle$, and $\overrightarrow{AB}$ is parallel to the direction vector $\mathbf{v} = \langle 1, 2, 1 \rangle$. So, if we placed $\mathbf{v}$ so that its initial point is at A it would run in the same direction or the exact opposite direction of $\overrightarrow{AB}$. This means that, by definition:

$$\|\overrightarrow{AB}\| = |\text{comp}_{\mathbf{v}}(\mathbf{u})| = \left| \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|} \right| = \left| \frac{6}{\sqrt{6}} \right| = \sqrt{6}.$$

Since $\|\mathbf{u}\| = \sqrt{17}$, we know by the Pythagorean theorem that $17 = d^2 + 6 \Rightarrow d = \sqrt{11}$ which is the answer.

Note: For another interesting way to do this problem example see question 30 of section 1.4 from the recommended homework.